Mathematical Induction

Questions 1–79

1.46 — Question 1. Show that

a) $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} = \begin{pmatrix} 2(k+1)+1 & -4(k+1) \\ k+1 & 1-2(k+1) \end{pmatrix}$

b) $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2^k-1 \\ 0 & 2^k \end{pmatrix} = \begin{pmatrix} 1 & 2^{k+1}-1 \\ 0 & 2^{k+1} \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^k-1) & 5^k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1} \end{pmatrix}$

1.46 — Question 2. Prove by induction, for all integers $n \geq 1$, that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.

1.46 — Question 3. Given $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, prove for all integers $n \geq 1$ that $A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$.

1.46 — Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -2n \\ 2n & 1-2n \end{pmatrix}$.

1.46 — Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & 2i \\ 2i & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & 2ni \\ 2ni & 1-2n \end{pmatrix}$.

1.46 — Question 6. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}^n = \begin{pmatrix} 2^n & 4^n - 2^n \\ 0 & 4^n \end{pmatrix}$.

1.46 — Question 7. Given that $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$.

1.46 — Question 8. Given that $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) \\ 0 & 3^n \end{pmatrix}$.

1.46 — Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}^n = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 1-4^n & 4^n \end{pmatrix}$.

1.46 — Question 10. Given that $A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$,

a) Find A^2, A^3 and A^4.
b) Write down a suitable form for A^n where n is a positive integer.
c) Prove your answer to part b).

1.46 — Question 11. Given that $A = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}$ where a is real, prove $A^n = \begin{pmatrix} 4^n & \frac{a}{3}(4^n - 1) \\ 0 & 1 \end{pmatrix}$ is true for all positive integers n.

1.46 — Question 12. Given that $A = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}$,

a) Prove that for $n \in \mathbb{N}$, $A^n = \begin{pmatrix} 1 - 3n & -9n \\ n & 3n + 1 \end{pmatrix}$.

b) Find the line of invariant points of the transformation represented by matrix A.

c) Show, algebraically, that the transformation A^n has the same line of invariant points.

1.46 — Question 13. It is known that $A = \begin{pmatrix} 2 & -a \\ a & 2 \end{pmatrix}$ where a is imaginary and $\det A = 0$.

a) Find the value of a.

b) Prove by mathematical induction, for $n \in \mathbb{N}$, $A^n = 4^{n-1}A$.

1.46 — Question 14. Two square matrices, A and B, are such that $AB = B^4A$ and $B^5 = I$. Prove that $B^nAB^n = A$ for $n \in \mathbb{N}$.

1.46 — Question 15. Prove by mathematical induction, for $n \in \mathbb{N}$, that

a) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 2n & \frac{1}{2}n(n+3) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 1-2^n & 3(2^n-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

1.47 — Question 1. Show that each of the following are divisible by 4.

a) $(2n+1)(2n-3) - 4n + 23$

b) $16^{3k} - 4$

c) $2^{k+2} - 4 \times 3^k + 8$

1.47 — Question 2. Show that each of the following are divisible by 6.

a) $6^{k+1} - 18$

b) $2^{k+1} \times 3^{k+2}$

c) $(k^2 + 1)(k - 4) + 10(k^2 + k + 1)$

1.47 — Question 3. Given that a, b and c are positive integers, decide whether each of the following statements are true or false:

a) If a and b are factors of c then $a + b$ is a factor of c.

b) If a is a factor of both b and c then a is a factor of $b + c$.

c) If a is a factor of $b + c$ then both b and c are factors of a.

d) If a is a factor of b and b is a factor of c then a is a factor of $b + c$.

1.47 — Question 4. Prove, by induction, that $n^2 + n$ is divisible by 2 for all integers $n \geq 1$.

1.47 — Question 5. Using proof by induction, prove that

a) $n^3 - n$ is divisible by 3

b) $n^3 + 3n^2 + 2n$ is divisible by 6 for all integers $n \geq 1$

1.47 — Question 6. Prove, by induction, that $9^n + 7$ is divisible by 8 for all integers $n \geq 1$.

1.47 — Question 7. Using proof by induction, prove that $5^n - 4n + 3$ is divisible by 4 for all integers $n \geq 0$.

1.47 — Question 8. Using proof by induction, prove that $5^n + 4n - 1$ is divisible by 8 for all integers $n \geq 1$.

1.47 — Question 9. Prove, by induction, that $6^n - 2^n$ is a multiple of 4 for all integers $n \geq 1$.

1.47 — Question 10. Prove, by induction, that

a) $5^n + 3 \times 9^n$ is divisible by 4

b) $5^{2n} + 3^{2n-1}$ is divisible by 4 for all integers $n \geq 1$

1.47 — Question 11. Prove, by induction, that $15^n - 8^{n-2}$ is divisible by 7 for all integers $n \geq 2$.

1.47 — Question 12. Prove, using induction, that for all positive odd integers n, $3^n + 7^n$ is divisible by 10.

1.47 — Question 13. The following shows a student's answer to the question "Prove, by induction, that $3^{2n} - 1$ is a multiple of 8 for all integers $n \geq 1$".

> True for $n = 1$
> When $n = k + 1$
> $(3^{2(k+1)} - 1) - (3^{2k} - 1)$
> $= 3^{2k} \times 3^2 - 3^{2k} = 3^{2k}(9 - 1) = 8 \times 3^{2k}$
> which is a multiple of 8 so true for all $n \geq 1$

a) Find all the errors.
b) Correct the proof.

1.47 — Question 14. Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 3.

1.47 — Question 15. For $n \geq 1$, it is given that $f(n) = 50n^2 + 70n$.

a) Show that $f(n+1) - f(n) = 20(5n+6)$.

b) Hence prove by induction that $f(n)$ is divisible by 4 for all integers $n \geq 1$.

1.47 — Question 16. For $n \geq 1$, it is given that $f(n) = 24 \times 2^{4n} + 3^{4n}$.

a) Find $f(n+1) - f(n)$.

b) Hence prove by induction that $f(n)$ is divisible by 5 for all integers $n \geq 1$.

1.47 — Question 17.

a) Prove that for all integers $n \geq 1$, $f(n) = 2n^3 - 5n$ is a multiple of 3.

b) Show that $f(-x) = -f(x)$ for all real values of x.

c) Hence prove that $f(n)$ is a multiple of 3 for all integer values of n.

1.48 — Question 1. Show that

a) $\frac{1}{3}k(k+1)(k+2)+(k+1)(k+2)=\frac{1}{3}(k+1)(k+2)(k+3)$

b) $\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}=\frac{k+1}{k+2}$

c) $(k+1)!-1+(k+1)(k+1)!=(k+2)!-1$

1.48 — Question 2. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$

b) $\sum_{r=1}^{n} 4^r = \frac{4}{3}(4^n - 1)$

1.48 — Question 3. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$

b) $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$

c) $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$

1.48 — Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n-1)(n+1)$.

1.48 — Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, $-3-4-3+\cdots+n(n-4) = \frac{1}{6}n(n+1)(2n-11)$.

1.48 — Question 6. Michael is partway through a proof, he writes: $\sum_{r=1}^{k+1} r(r+3) = \sum_{r=1}^{k} r(r+3) + (k+1)$. Identify his mistake and correct this line of working.

1.48 — Question 7. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)} = \frac{n}{2n+4}$.

1.48 — Question 8. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} = \frac{n(7n+25)}{24(n+3)(n+4)}$.

1.48 — Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} \frac{1}{3^r} = \frac{1}{2} - \frac{1}{2 \times 3^n}$

b) $\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$

c) $\sum_{r=1}^{n} (r+1)2^r = n \times 2^{n+1}$

1.48 — Question 10. It is conjectured that for constants a, b and c, $S_n = \sum_{r=1}^{n} (r+1)^2 = \frac{1}{6} n(an^2 + bn + c)$.

a) By considering S_1, S_2 and S_3 show that $a = 2$, $b = 9$ and $c = 13$.

b) Prove the conjecture by induction.

1.48 — Question 11. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n}(r+i) = -n + \frac{1}{2}n(n+1)i$.

1.48 — Question 12. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r \times r! = (n+1)! - 1$.

1.48 — Question 13. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$.

1.48 — Question 14. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)} = \frac{n}{(1+a)(n+1+a)}$.

1.48 — Question 15. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} a^r = \frac{a(a^n - 1)}{a - 1}$.

1.49 — Question 1. Show that

a) $3\left(3^{k-1} + \frac{1}{2}\right) - 1 = 3^k + \frac{1}{2}$

b) $3(2^k + 3^k) - 2^k = 2^{k+1} + 3^{k+1}$

c) $\frac{\frac{3}{3k-1}}{1 + \frac{3}{3k-1}} = \frac{3}{3(k+1) - 1}$

d) $\frac{1}{2}(k^2 - 5k + 12) + k - 2 = \frac{1}{2}((k+1)^2 - 5(k+1) + 12)$

1.49 — Question 2. Given that $u_1 = 8$ and $u_{n+1} = u_n - 3$, prove that for all integers $n \geq 1$, $u_n = 11 - 3n$.

1.49 — Question 3. Given that $u_1 = 2$ and $u_{n+1} = 2u_n - 1$, prove by mathematical induction for $n \in \mathbb{N}$ that $u_n = 2^{n-1} + 1$.

1.49 — Question 4. Given that $u_1 = 8$ and $u_{n+1} = 4u_n + 3$, prove that for all integers $n \geq 1$, $u_n = 9 \times 4^{n-1} - 1$.

1.49 — Question 5. Given that $u_1 = 4$ and $u_{n+1} = 3u_n - 1$, prove by mathematical induction for all integers $n \geq 1$ that $u_n = \frac{1}{6}(7 \times 3^n + 3)$.

1.49 — Question 6. Given that $u_1 = 2$ and $u_{n+1} = 8u_n + 5$, prove that for all integers $n \geq 1$, $u_n = \frac{19 \times 8^n - 40}{56}$.

1.49 — Question 7. Given that $u_1 = 2$ and $u_{n+1} = u_n + 3n + 5$, prove by mathematical induction for $n \in \mathbb{N}$ that $u_n = \frac{1}{2}(3n^2 + 7n - 6)$.

1.49 — Question 8. Given that $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1+u_n}$, prove that for all integers $n \geq 1$, $u_n = \frac{1}{n}$.

1.49 — Question 9. Given that $u_1 = 2$ and $u_{n+1} = \dfrac{2u_n + 1}{u_n + 2}$, prove that for all integers $n \geq 1$, $u_n = \frac{3^n+1}{3^n-1}$.

1.49 — Question 10. A sequence $z_1, z_2, z_3, \ldots, z_n$ of complex numbers is defined recursively as follows: $z_1 = 1 + i$ and $z_{n+1} = (1 + i)z_n + (1 - i)^n$ for all integers $n \geq 1$.

a) Find z_2 and z_3 expressing your answers in the form $a + bi$ where a and b are real numbers.

b) Prove by mathematical induction that $z_n = \frac{1}{2}(1 + i)^n + \frac{i}{2}(1 - i)^n$ for all integers $n \geq 1$.

1.49 — Question 11. A sequence $u_1, u_2, u_3, \ldots, u_n$ is defined by the recurrence relation $u_1 = 3$, $u_{n+1} = \dfrac{u_n}{2 + u_n}$.

a) Find u_2, u_3 and u_4.

b) Find the first 4 terms of $2^{n+1} - 3$.

c) Suggest an expression for u_n in terms of n.

d) Prove by mathematical induction that your expression for u_n is correct.

1.49 — Question 12. Given that $u_1 = q$ and $u_{n+1} = u_n + a$ for constants q and a, prove by mathematical induction for all integers $n \geq 1$ that $u_n = q + (n-1)a$.

1.49 — Question 13. Given that $u_1 = 5$, $u_2 = 13$ and $u_{n+2} = 3u_{n+1} - 2u_n$,

a) Write down the values of u_3 and u_4.

b) Prove that for all integers $n \geq 1$, $u_n = -3 + 2^{n+2}$.

1.50 — Question 1. Simplify each of the following:

a) 2×2^k

b) -3×3^k

c) $a \times a^k$

1.50 — Question 2. Simplify each of the following:

a) $(1+a) \times (1+a)^k$

b) $(1-a) \times (1-a)^k$

1.50 — Question 3. Simplify each of the following:

a) $3 \times 2!$

b) $12 \times 11!$

c) $(k+1)k!$

1.50 — Question 4. Use proof by induction to show that $2^n > 2n$ for all positive integers n such that $n \geq 3$.

1.50 — Question 5. Use mathematical induction to prove that $4^n > 3n + 7$ for all integers $n \geq 2$.

1.50 — Question 6. Prove by induction, for all integers $n \geq 5$, that $n^2 < 2^n$.

1.50 — Question 7. It is given that $n^3 < 3^n$ where n is a positive integer and a is the largest value of n for which the inequality does not hold true.

a) Find the value of a.

b) Show that it is true for $n < a$.

c) Prove, using mathematical induction, that for all $n > a$, $n^3 < 3^n$ holds true.

1.50 — Question 8. Use proof by induction to show that $n! > 2^n$ for all positive integers n such that $n \geq 4$.

1.50 — Question 9. Prove by induction, for all integers $n \geq 5$, that $n! > 10n$.

1.50 — Question 10. Prove by induction, for all integers $n \geq 5$, that $3^n > (n+1)^3$.

1.50 — Question 11. Prove by induction that $n! > n^2 + n$ for all integers $n \geq 4$.

1.50 — Question 12. Using mathematical induction, prove that $2 > \left(1 + \frac{1}{n}\right)^3$ for all integers $n \geq 4$.

1.50 — Question 13. For real number a and integer number n where $a > -1$ and $n \geq 2$, prove by mathematical induction that $(1+a)^n > 1 + an$.

1.50 — Question 14. Prove by induction, for all integers $n \geq 2$, that $n! < n^n$.

1.50 — Question 15. Given that A and B are non-singular square matrices of equal dimensions. Use proof by induction to show that $(ABA^{-1})^n = AB^nA^{-1}$ for all integers $n \geq 1$.

1.50 — Question 16. It is known that A is a square matrix such that $A^2 = A$. Use proof by induction to show that for all positive integers n, $I + A + A^2 + \cdots + A^n = I + nA$.

1.50 — Question 17. It is known that A is a square matrix and a sequence of matrices is defined recursively as $A_1 = A$ and $A_{n+1} = A \times A_n$. Use proof by induction to show that for all positive integers n, $\det(A_n) = (\det(A))^n$.

1.50 — Question 18. Use proof by induction to show that $\dfrac{d^n}{dx^n}(x^a) = \dfrac{a!}{(a-n)!}x^{a-n}$ for all integers $n \geq 1$ where $n \leq a$.

1.50 — Question 19. Prove using mathematical induction that if there are n lines in a plane, no two of which are parallel and no three meet at a point, then they divide the plane into $\frac{n^2+n+2}{2}$ regions.